

Post-IPO Performance Analysis and SpaceX (SPCX) Valuation Proxy Study

A quantitative review of twelve high-profile post-2019 IPOs, used as historical proxies to assess the risk profile of an anticipated SpaceX public listing.

1. Introduction & Methodology.....	3
2. IPO Pop: First-Day Price Reaction	3
3. Multiple Compression: Price-to-Sales Ratio Over Time	5
4. Risk-Adjusted Returns: Sharpe Ratio	6
5. Stress Test: Maximum Drawdown in the First 24 Months	7
6. Normalised Price Trajectories	8
7. Directional Matrix: Hype vs. Long-Run Quality	9
8. SpaceX Fundamentals: A Standalone Valuation Model	9
8.1 2025 Revenue Composition.....	9
8.2 Growth Projections to 2030.....	10
8.3 Headline Valuation Metrics	11
9. Predictive Model: P/S Ratio vs. Drawdown Risk.....	11
10. Stochastic Simulation: Multiple Decay	12
11. Summary & Key Takeaways	13

1. Introduction & Methodology

This report analyses how twelve tech and “new economy” companies have performed since their IPOs between 2019 and 2023: Rivian (RIVN), Palantir (PLTR), Snowflake (SNOW), Rocket Lab (RKLB), Arm (ARM), Mobileye (MBLY), Coinbase (COIN), AST SpaceMobile (ASTS), Virgin Galactic (SPCE), Planet Labs (PL), BlackSky (BKSX) and Redwire (RDW).

These companies share a lot of DNA with SpaceX (let’s call it SPCX): huge pre-IPO hype, big exposure to space, mobility, AI or deep-tech, super premium revenue multiples at the start, and tons of retail investor interest. They’re basically our “proxy basket” to model how SpaceX might behave once it hits the public markets.

The analysis covers eight dimensions:

- **IPO Pop:** The percentage jump from the issue price to the first day's opening trade.
- **Price-to-Sales (P/S) multiple compression:** How valuation multiples shake out relative to revenue over time.
- **Sharpe Ratio:** Annualised returns since listing, risk-adjusted using a 4.2% risk-free rate.
- **Maximum Drawdown:** The worst peak-to-trough drop within 24 months, translated into a dollar loss for a \$1,780 bn SpaceX valuation.
- **Normalised price trajectories:** All stocks rebased to 100 at listing to compare their relative paths.
- **A directional matrix:** Cross-referencing initial hype (IPO Pop) against long-run risk-adjusted return (Sharpe Ratio).
- **SpaceX fundamentals:** A standalone valuation model based on 2025 revenues and Goldman Sachs' growth projections to 2030.
- **A predictive model:** A regression model and a stochastic simulation to forecast how SpaceX's multiple might evolve after listing.

All price data is from Yahoo Finance, and fundamentals come from quarterly income statements. Treat this as an illustrative, model-driven exercise, not a crystal ball—actual outcomes for SpaceX will depend on market conditions, lock-up structures, and the financials they disclose when they list.

2. IPO Pop: First-Day Price Reaction

The “IPO Pop” is just the percentage gain or loss between the underwriter's offering price and the first day's opening trade. A massive pop usually means the IPO was way too cheap compared to immediate market demand—it's often a sign that retail hype is getting ahead of the actual fundamentals.

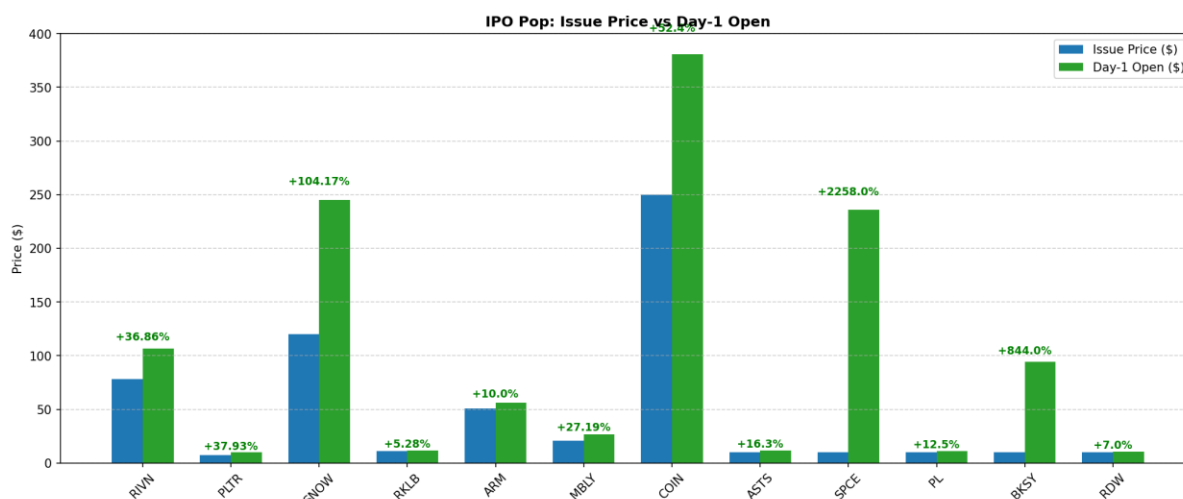


Figure 1 – Issue price vs. Day-1 opening price, with the resulting pop percentage.

Ticker	Issue Price (\$)	Day-1 Open (\$)	IPO Pop (%)
RIVN	78.00	106.75	+36.86%
PLTR	7.25	10.00	+37.93%
SNOW	120.00	245.00	+104.17%
RCLB	11.00	11.58	+5.28%
ARM	51.00	56.10	+10.00%
MBLY	21.00	26.71	+27.19%
COIN	250.00	381.00	+52.40%
ASTS	10.00	11.63	+16.30%
SPCE	10.00	235.80	+2,258.00%
PL	10.00	11.25	+12.50%
BKSJ	10.00	94.40	+844.00%
RDW	10.00	10.70	+7.00%

Most of the basket shows moderate pops in the 5-50% range, which is pretty standard. However, three names stand out:

- Virgin Galactic (SPCE): +2,258% — this was driven by a SPAC structure, not a traditional IPO, where the pre-merger price had already detached from the \$10 trust value.
- BlackSky (BKSJ) and Redwire (RDW): Similar SPAC listings with outsized first-day moves for the same structural reasons.
- Snowflake (SNOW): +104% remains the high-water mark for traditional book-built IPOs, reflecting the absolute mania for high-growth SaaS in 2020.

If SpaceX goes the traditional IPO route—which is the consensus—the better comparisons are SNOW, COIN, RIVN and PLTR, which posted pops between roughly +37% and +104%. This suggests a SpaceX IPO, if priced near its private valuation, could realistically open 30-100% higher on day one if the demand holds up.

3. Multiple Compression: Price-to-Sales Ratio Over Time

The Price-to-Sales (P/S) ratio measures how many dollars of market value investors are willing to pay for every dollar of annualised revenue. Newly listed growth stocks usually start with eye-watering multiples that then “mean-revert” to more sustainable levels as the market starts valuing the business on cold, hard numbers rather than the growth story.

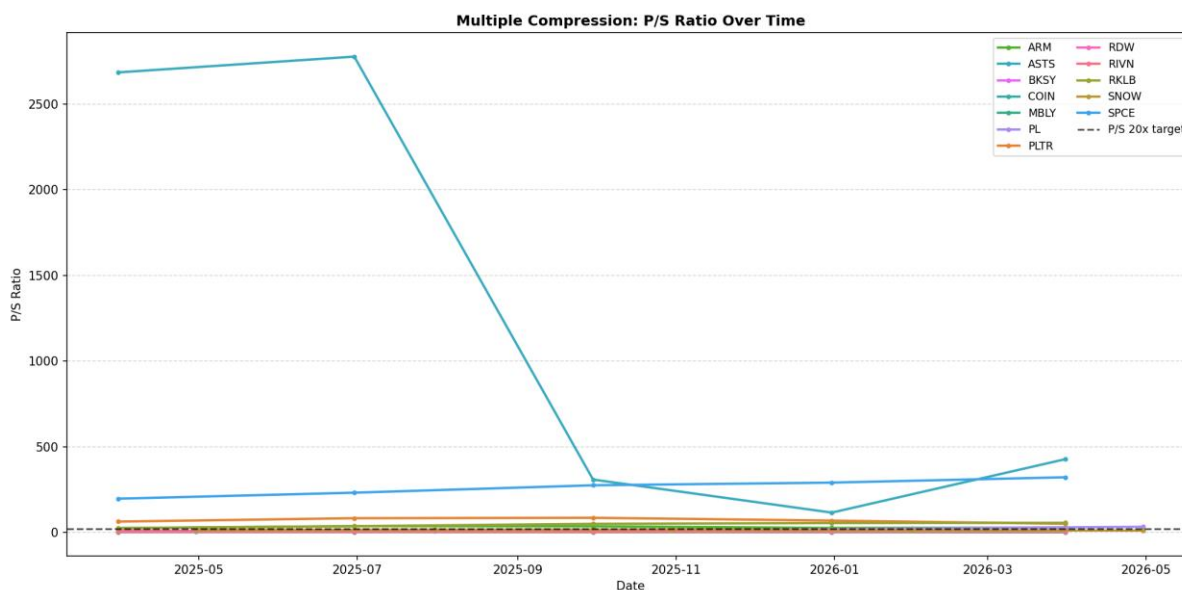


Figure 2 – Quarterly P/S ratio for each ticker, against a 20x reference line.

The chart is totally skewed by AST SpaceMobile (ASTS), which hit a peak of roughly 2,780x in mid-2025—just because its commercial revenue was basically zero—before compressing down to 100-430x as the business scaled. Excluding ASTS, the rest of the pack trades in a much tighter 1x-90x range:

Ticker	First Observed P/S	Latest P/S
ARM	25.41x	28.53x
ASTS	2,684.87x	427.61x
BKSY	3.18x	13.26x
COIN	5.48x	7.43x
MBLY	2.08x	0.90x
PL	4.81x	32.82x
PLTR	63.11x	50.36x
RDW	9.52x	4.92x
RIVN	3.32x	3.76x
RKLB	24.21x	56.06x
SNOW	14.20x	10.60x
SPCE	196.77x	321.88x

Palantir (PLTR) and Virgin Galactic (SPCE) show how the AI/space narrative can sustain—or even increase—elevated multiples, with PLTR staying well above the 20x reference line. On the flip side, Mobileye (MBLY), Rivian (RIVN) and Redwire (RDW) have compressed to single-digit multiples, looking more like mature industrial or automotive stocks.

For SpaceX, this is the kicker: the modelled P/S ratio of 95.2x puts it at the very top of this historical range—comparable only to the early days of ASTS and the AI-driven re-rating of PLTR. This implies SpaceX would be priced for perfection from day one; if growth slows down, there's a massive amount of room for that multiple to get crushed.

4. Risk-Adjusted Returns: Sharpe Ratio

The Sharpe ratio tells you how much return you're getting for the amount of volatility you're taking on (using a 4.2% risk-free rate). Anything above 1.0 is a great risk-adjusted performance; near or below zero means the volatility just isn't being rewarded.

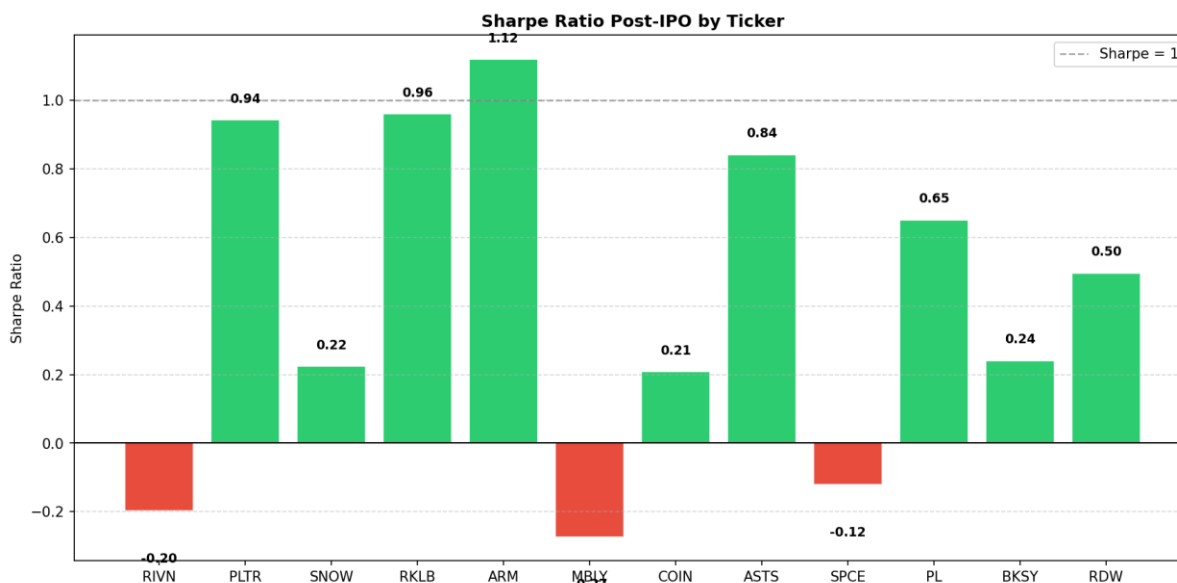


Figure 3 – Annualised Sharpe ratio since listing, by ticker.

Ticker	Annual Return (%)	Annual Volatility (%)	Sharpe Ratio
RIVN	-11.16%	77.49%	-0.198
PLTR	70.00%	69.76%	0.943
SNOW	18.26%	62.69%	0.224
RKLB	82.43%	81.42%	0.961
ARM	89.29%	76.00%	1.120
MBLY	-12.29%	60.41%	-0.273
COIN	22.05%	85.34%	0.209
ASTS	97.56%	110.87%	0.842
SPCE	-7.82%	99.58%	-0.121
PL	59.45%	84.85%	0.651

BKSY	28.73%	101.68%	0.241
RDW	52.08%	96.50%	0.496

Three companies—Rivian (-0.20), Mobileye (-0.27) and Virgin Galactic (-0.12)—posted negative Sharpe ratios. That means investors who bought at the IPO have actually been worse off than holding a risk-free asset, despite (in Rivian’s case) dealing with massive volatility.

At the other end, Arm (1.12), Rocket Lab (0.96) and Palantir (0.94) all cleared the 1.0 hurdle, even with annual volatility over 70-80%. This shows that high volatility doesn't kill you if the returns are persistent. For a SpaceX listing, the takeaway is that success doesn't hinge on the initial hype, but on whether they can execute and grow over the first 12-24 months.

5. Stress Test: Maximum Drawdown in the First 24 Months

This is the big risk warning. We looked at the worst peak-to-trough decline for each stock within their first two years.

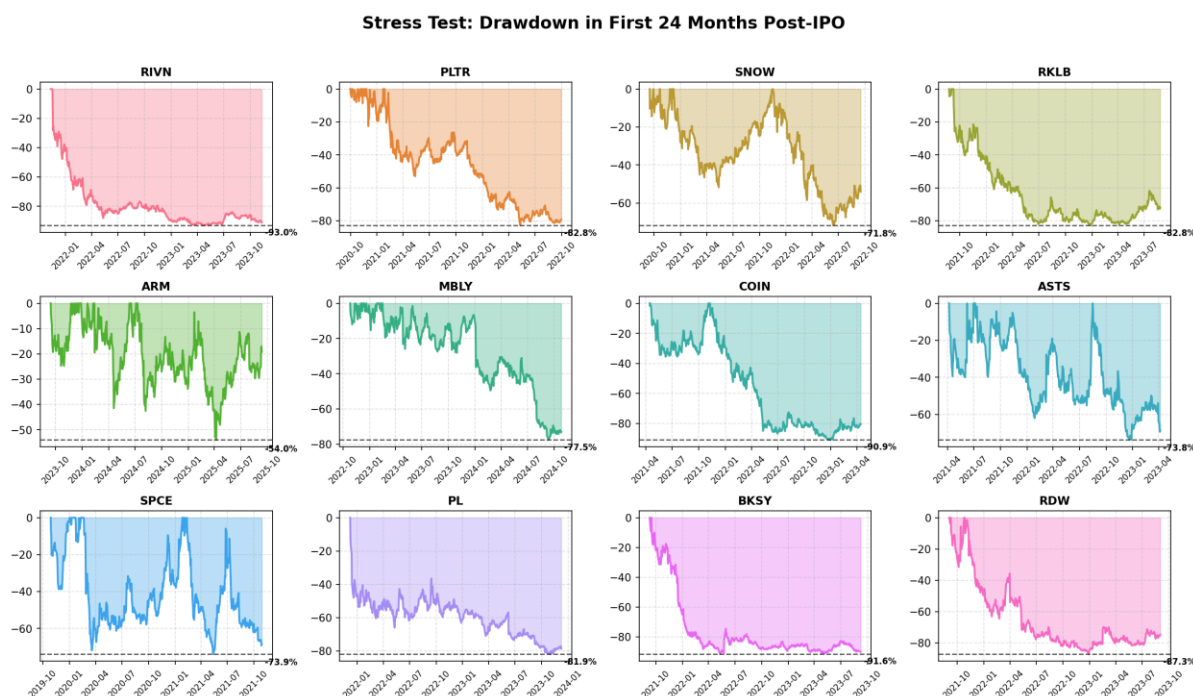


Figure 4 – Drawdown profiles (%) for each ticker over their first 24 months of trading.

Ticker	Max Drawdown (%)	Peak Date	Trough Date	Implied SpaceX Loss (\$bn)
RIVN	-93.02%	2021-11-16	2023-04-25	-1,656
PLTR	-82.79%	2021-01-27	2022-05-11	-1,474
SNOW	-71.81%	2021-11-16	2022-06-13	-1,278
RKLK	-82.82%	2021-09-09	2022-12-27	-1,474
ARM	-53.97%	2024-07-10	2025-04-08	-961
MBLY	-77.52%	2023-02-15	2024-09-12	-1,380

COIN	-90.90%	2021-11-09	2022-12-28	-1,618
ASTS	-73.80%	2022-08-16	2022-12-27	-1,314
SPCE	-73.91%	2021-02-11	2021-05-13	-1,316
PL	-81.85%	2021-12-08	2023-11-01	-1,457
BKSY	-91.56%	2021-09-16	2022-05-11	-1,630
RDW	-87.26%	2021-10-26	2022-12-28	-1,553

Every single company in our group had a drawdown of more than 50% within two years of listing, and ten of the twelve fell by over 70%. The average drawdown across the whole group is roughly -80%. If you apply that to a \$1,780 bn SpaceX valuation, you're looking at a paper loss of around \$1,450-1,500 bn from the peak. Regardless of how strong the business is, almost every peer here hit a wall in the first two years due to macro conditions, lock-up expirations, and the inevitable return to normal valuation multiples.

6. Normalised Price Trajectories

Rebasing every stock to 100 on the day they list lets us compare how they actually performed over time, regardless of share price.

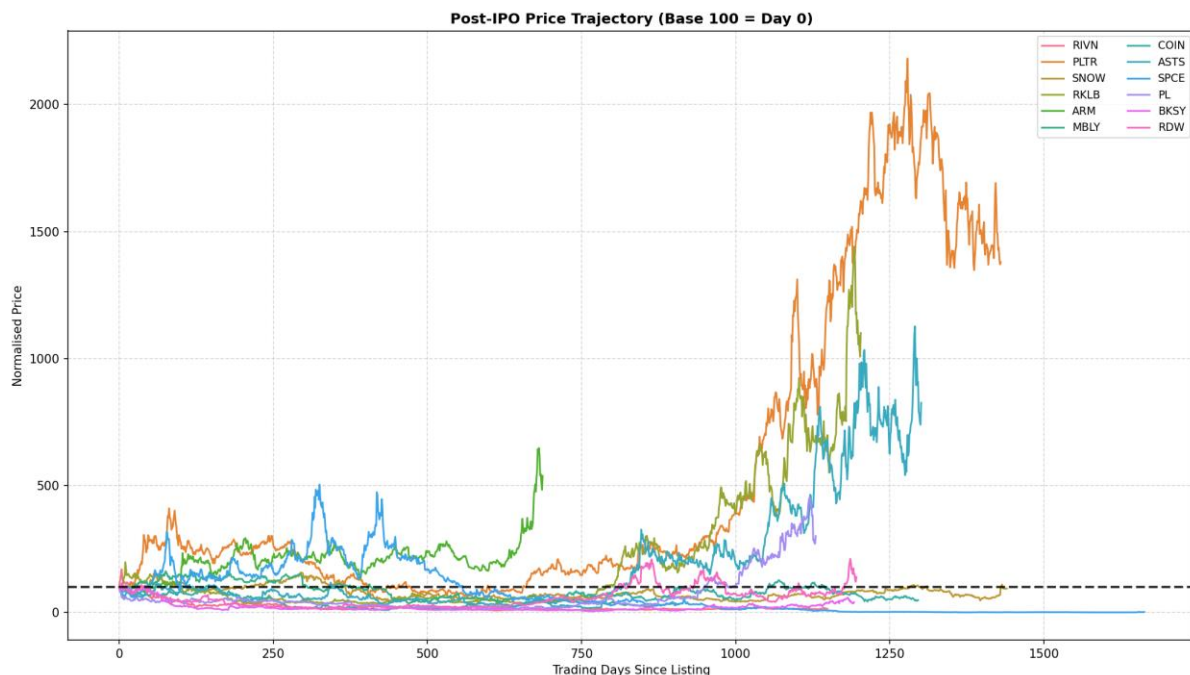


Figure 5 – Post-IPO price trajectories, indexed to 100 at Day 0.

The winners—Palantir (PLTR) and Rocket Lab (RKL)—spent their first 2-3 years trading sideways or below their IPO price before finally catching a bid around 800-1,000 trading days later, once their AI/space narratives started paying off. Most of the rest, like SPCE, RIVN, BKSY and others, spent most of their time below the starting line. For SpaceX, the lesson is that even the great success stories usually force investors to sit through a long, painful period of underperformance before things get better.

7. Directional Matrix: Hype vs. Long-Run Quality

This chart plots the initial IPO Pop against the long-term Sharpe ratio.

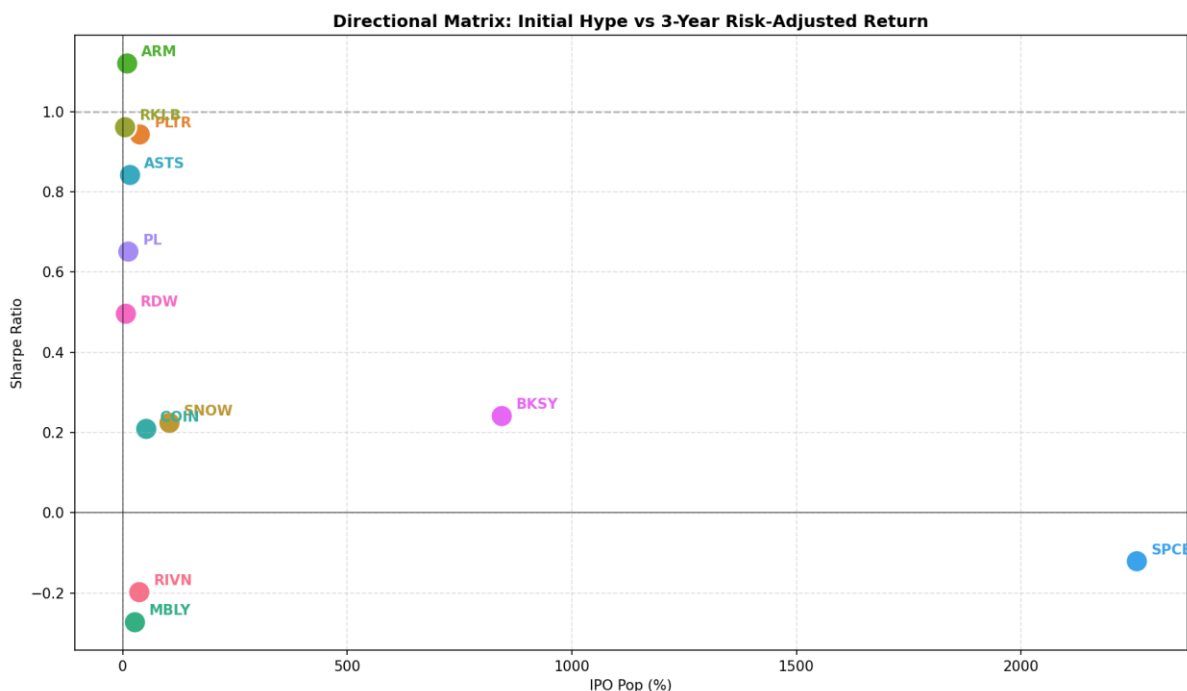


Figure 6 – IPO Pop (%) vs. Sharpe Ratio, by ticker.

There's basically zero correlation here—if anything, it's slightly negative. The companies with the biggest pops (SPCE, BKSJ) ended up being some of the worst long-term performers. Meanwhile, the three best stocks (ARM, RKL, PLTR) all had modest pops below 40%. A big first-day jump isn't a signal of quality; if anything, it's a warning sign of pure speculation.

8. SpaceX Fundamentals: A Standalone Valuation Model

8.1 2025 Revenue Composition

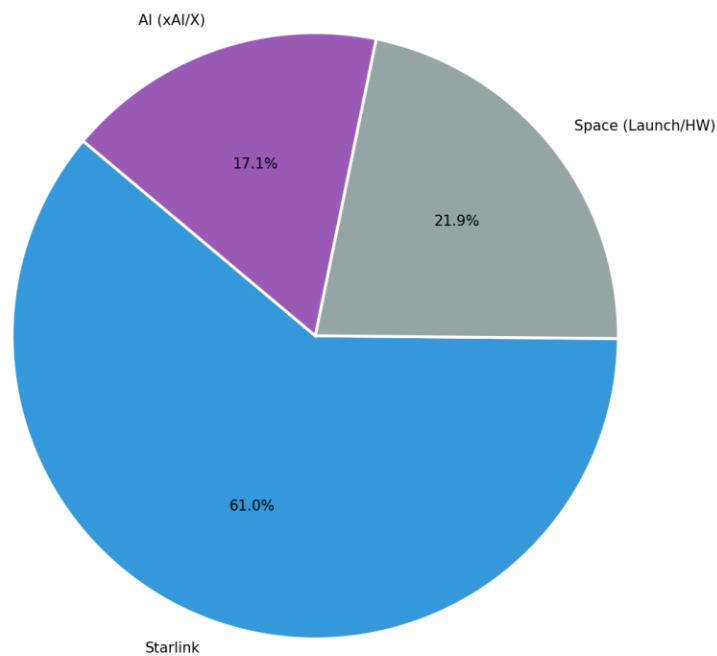
SpaceX 2025 Revenue Breakdown (Total: \$18.7 bn)

Figure 7 – SpaceX 2025 revenue breakdown by segment (total: \$18.7 bn).

Starlink is currently the anchor, making up 61% of 2025 revenue (\$11.4 bn), with Space launch/hardware at 21.9% (\$4.1 bn) and the new AI segment (xAI/X) at 17.1% (\$3.2 bn).

8.2 Growth Projections to 2030

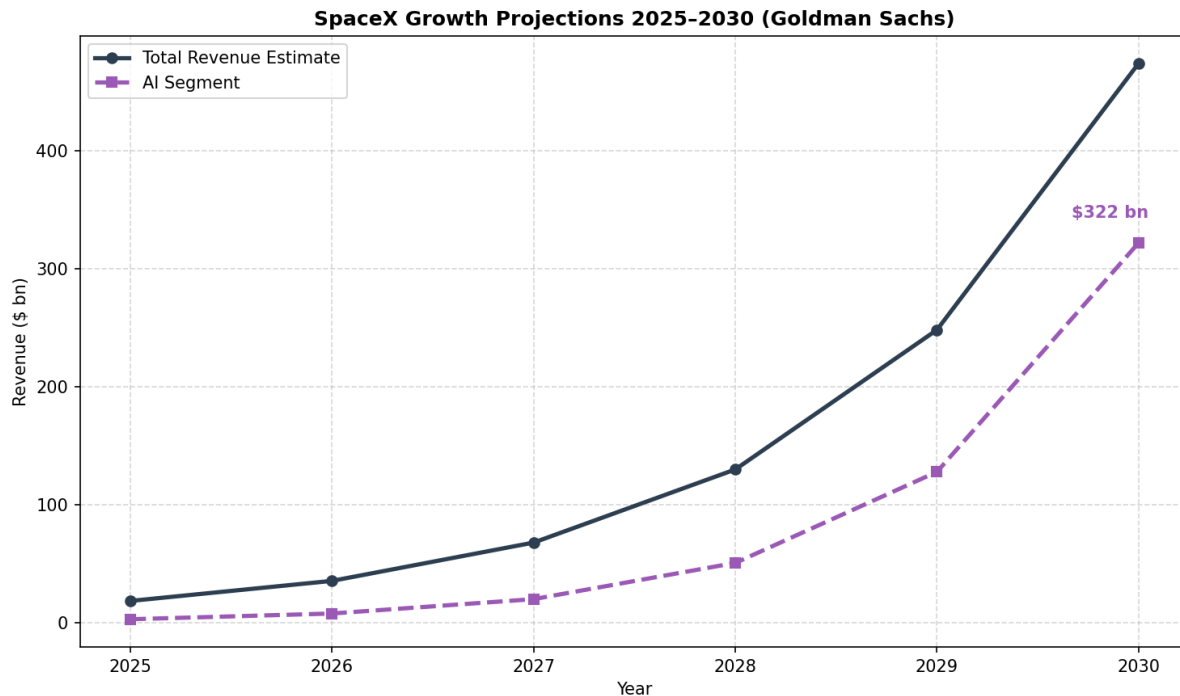


Figure 8 – Goldman Sachs total revenue and AI-segment projections, 2025-2030.

According to Goldman Sachs, revenue needs to grow from \$18.7 bn in 2025 to \$474 bn by 2030, with the AI segment alone going from \$3.2 bn to \$322 bn over the same period—implying an AI revenue CAGR of roughly 151.5% versus a total-company CAGR of about 88%. By 2030, AI would represent nearly 68% of total projected revenue.

8.3 Headline Valuation Metrics

Metric	Value
Target Valuation (equity)	\$1,780 bn
Enterprise Value	\$1,778 bn
Cash	\$25 bn
Debt	\$23 bn
Total Revenue 2025	\$18.7 bn
P/S Ratio 2025	95.2x
EV/Sales 2025	95.1x
AI Revenue CAGR Required (2025-2030)	151.5%

At a \$1,780 bn valuation, you're paying 95x sales—a multiple that is, with the exception of that early ASTS spike, totally unprecedented. SpaceX would be entering the public markets priced for a growth trajectory that almost nobody else in this basket has ever managed to sustain.

9. Predictive Model: P/S Ratio vs. Drawdown Risk

Running a regression across the basket suggests that with a 95.2x P/S multiple, SpaceX is looking at a max drawdown in the 75-85% range.

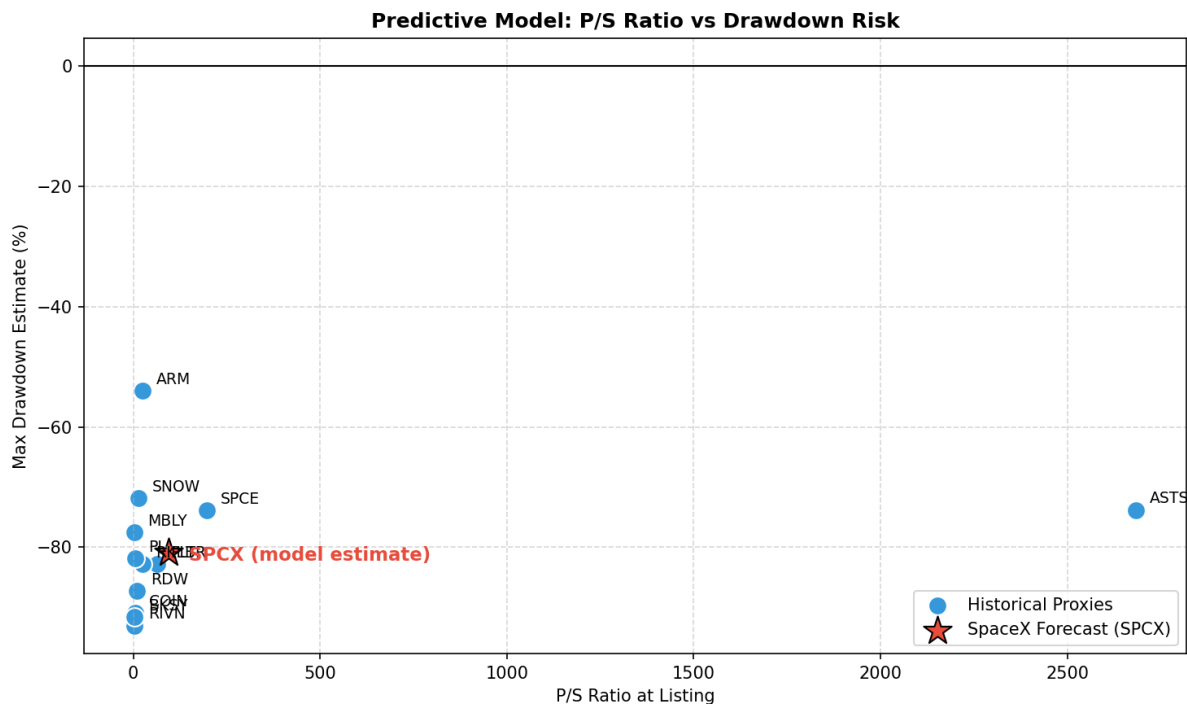


Figure 9 – Historical P/S ratio at listing vs. realised max drawdown, with the SpaceX (SPCX) model estimate highlighted.

This aligns with the rest of the group. A peak-to-trough decline of \$1,330-1,510 bn should be viewed as the “historical norm” for this kind of premium valuation, not a tail risk.

10. Stochastic Simulation: Multiple Decay

To see how that 95.2x multiple might settle, we ran a mean-reversion simulation. The model assumes the multiple drifts toward a more sustainable 20x target over 24 months.

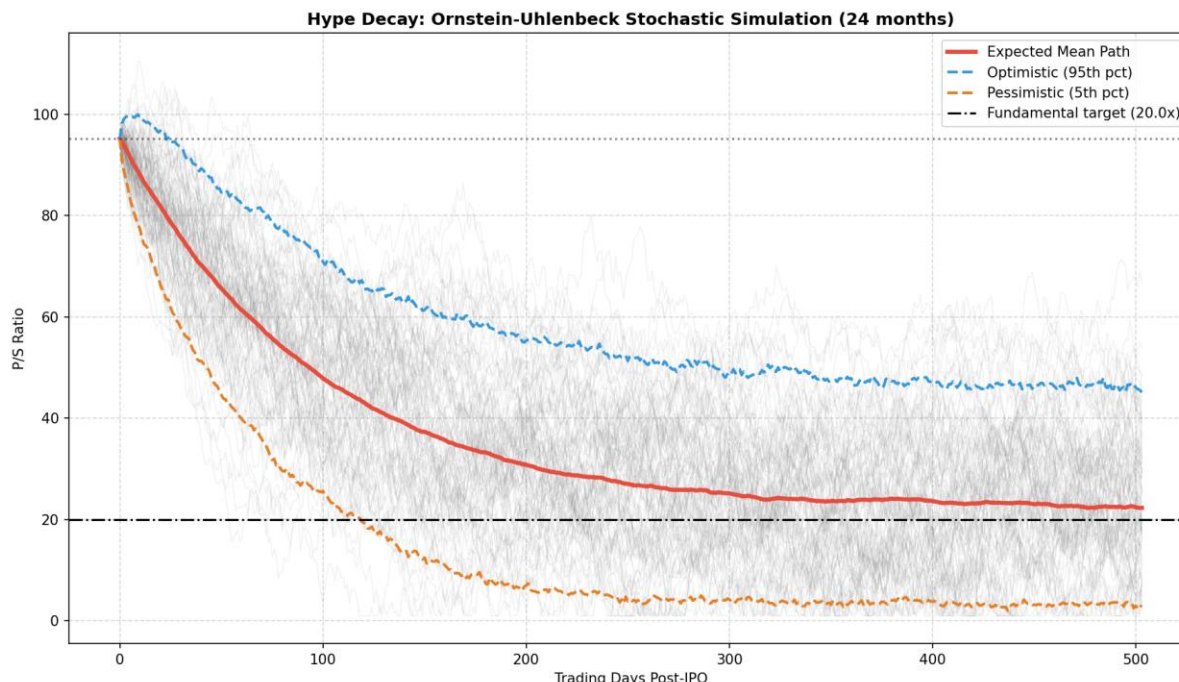


Figure 10 – Simulated P/S ratio paths starting from 95.2x, with mean, 5th and 95th percentile bands, and the 20x fundamental target.

The expected path is a drop from 95x to about 22-23x—a 76% compression in the multiple. It's important to remember this isn't necessarily a 76% crash in the share price, because strong revenue growth could offset some of it. But it shows the multiple has very little room to do anything but come back to earth.

11. Summary & Key Takeaways

- IPO Pop: You could see +30% to +100%, but don't read it as a sign of long-term quality—it's often the opposite.
- Valuation: At \$1,780 bn, the 95x sales multiple is extreme and rarely lasts.
- Downside Risk: Every single peer in this group saw a >50% drawdown within 24 months (averaging -80%).
- Multiple Normalisation: Expect the P/S ratio to grind down toward the 20-25x area within two years.
- Long-run outcomes: 25% of the basket delivered massive 5-20x returns later on, but only after surviving a long, brutal period of early underperformance.

In short: a SpaceX IPO at this valuation would be entering the public markets at an extreme multiple, with a very high probability of a severe (-70% to -90%) drawdown before settling down. But, for those who can stomach the ride, there is a real precedent for massive long-term outperformance if the Starlink and AI revenues actually scale like the projections say they will.